

Tools:

- GitHub Copilot: <https://github.com/copilot> (create an account using your @info.uaic.ro email address)
- WSL: <https://learn.microsoft.com/en-us/windows/wsl/install> or a Linux distribution
- VirtualBox (optional): <https://www.virtualbox.org>
- Docker / Docker Desktop (optional): <https://www.docker.com>
- Visual Studio Code: <https://code.visualstudio.com/> (or any other IDE; eg. JetBrains)

To access the internal development server, please follow the instructions here:

https://docs.google.com/document/d/1L70rephVBh9r8_US4khX0_7zSalxcJJFJq31Nd86NCM

(Credits: Computer Networks Course, Conf. Dr. Andrei Panu)

You are not restricted to a specific programming language. We encourage diversity, so you can choose any programming language or IDE you are comfortable with. However, the lab examples will be presented using C/C++ and Python.

Ghid instalare WSL:

https://edu.info.uaic.ro/sisteme-de-operare/SO/lectures/Linux/Mini-ghid_instalare_WSL2.pdf

(user: so2026, pass: 20+26=46; **sau vezi linkul de mai sus de la WSL**)

Ghid instalare Linux Mint:

https://edu.info.uaic.ro/sisteme-de-operare/SO/lectures/Linux/Install_guide.pdf

(credits: Lect. Dr. Vidrascu Cristian, same user and pass)

Resources:

- pthreads: <https://man7.org/linux/man-pages/man7/pthreads.7.html>
- Google Sanitizers: <https://github.com/google/sanitizers>
- Linux udp(7) man page: <https://man7.org/linux/man-pages/man7/udp.7.html>
- Linux tcp(7) man page: <https://man7.org/linux/man-pages/man7/tcp.7.html>
- QUIC Working Group (IETF): <https://quicwg.org/>
- Cloudflare "The Road to QUIC" (accessible overview): <https://blog.cloudflare.com/the-road-to-quic/>
- quiche (Cloudflare, Rust/C): <https://github.com/cloudflare/quiche>
- ngtcp2 (C): <https://github.com/ngtcp2/ngtcp2>
- msquic (Microsoft, C): <https://github.com/microsoft/msquic>
- aioquic (Python): <https://github.com/aiortc/aioquic>
- RFC 9000 -- QUIC Transport Protocol: <https://www.rfc-editor.org/rfc/rfc9000>
- RFC 9001 -- Using TLS to Secure QUIC: <https://www.rfc-editor.org/rfc/rfc9001>
- RFC 9114 -- HTTP/3 (over QUIC): <https://www.rfc-editor.org/rfc/rfc9114>
- RFC 768 -- UDP specification (the entire spec is 3 pages): <https://www.rfc-editor.org/rfc/rfc768>
- RFC 9293 -- TCP specification (2022, replaces RFC 793): <https://www.rfc-editor.org/rfc/rfc9293>

- Beej's Guide to Network Programming (practical C examples):
<https://beej.us/guide/bgnet/>